1. Mechanics

Kinematics (linear motion)

$v = u + at$

$s = ut + 1/2\ at^2$

$v^2 = u^2 + 2as$

Average velocity: $v_{avg} = (u + v)/2$

Newton's Laws

$F = ma$

Weight: $W = mg$

Momentum: $p = mv$

Impulse: $I = F\Delta t = \Delta p$

Work, Energy, Power

Work: $W = F \times d \times \cos\theta$

Kinetic Energy: $KE = 1/2\ mv^2$

Potential Energy: $PE = mgh$

Power: $P = W/t = Fv$

Circular Motion

Centripetal acceleration: $a_c = v^2/r$

Centripetal force: $F_c = mv^2/r$

2. Gravitation

Newton’s Law of Gravitation: $F = G\, m1\, m2 / r^2$

Acceleration due to gravity: $g = G M / R^2$

Orbital velocity: $v = \sqrt{(GM/R)}$

Escape velocity: $v_e = \sqrt{(2GM/R)}$

3. Fluid Mechanics

Density: $\rho = m/V$

Pressure: $P = F/A$

Hydrostatic pressure: $P = \rho gh$

Archimedes' principle: Buoyant force = weight of displaced fluid

4. Oscillations & Waves

Simple harmonic motion: $x = A \sin(\omega t + \phi)$

Angular frequency: $\omega = 2\pi f = \sqrt{(k/m)}$

Time period: $T = 2\pi/\omega$

Wave speed: $v = f\,\lambda$

Frequency-wavelength relation: $v = \lambda\, f$

5. Thermodynamics

Ideal gas equation: $PV = nRT$

Work done: $W = P\Delta V$

First law: $\Delta U = Q - W$

Efficiency: $\eta = W/Q_{in}$

Heat: $Q = mc\Delta T$

6. Electrostatics

Coulomb's law: $F = k\, q1\, q2 / r^2$

Electric field: $E = F/q = k\, Q / r^2$

Electric potential: $V = k\, Q / r$

Capacitance: $C = Q/V$

7. Current Electricity

Ohm's law: $V = IR$

Resistance: $R = \rho\, L / A$

Power: $P = VI = I^2R = V^2/R$

Series: $R_s = R1 + R2 + \ldots$; Parallel: $1/R_p = 1/R1 + 1/R2 + \ldots$

8. Magnetism

Force on current-carrying wire: $F = B\, I\, L \sin\theta$

Magnetic field around wire: $B = \mu_0\, I / (2\pi\, r)$

Lorentz force: $F = q(v \times B)$

9. Optics

Lens formula: $1/f = 1/v - 1/u$

Magnification: $m = h_i / h_o = v / u$

Mirror formula: $1/f = 1/v + 1/u$

Refraction: $n = c/v$

10. Modern Physics

Photoelectric effect: $E = hf = \phi + KE_{max}$

Einstein’s mass-energy: $E = mc^2$

Radioactive decay: $N = N_0 e^{(-\lambda t)}$

Half-life: $t_{1/2} = 0.693 / \lambda$